Empirical Methods in Finance

Class 5 - Multivariate Time Series Analysis

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Objectives of the lecture

When we turn to a multivariate framework, it is still possible to use time series tools, but at the same time to start thinking in terms of explanatory variables.

The simplest multivariate model is the **VAR model** (Vector AutoRegressive model), which allows modeling the interaction between a set of variables.

In this lecture, we will study:

1. VAR models
2. Causality relationships that can exist between variables
3. Impulse response analysis
4. Variance decomposition

Table of Contents

VMA models

Causality Tests

Practical Application - Oil Volatility Shocks Transmission

Impulse Response Functions

Practical Application - US economy Macro Model

Pratical Application - Building Spillover Measures

Practical Application: Inflation Shocks and Fed Rates in a VAR

VMA models

VMA(q) Model

A vector moving-average model of order q, or VMA(q), is in the form

$$r_t = \mu + \varepsilon_t - \Theta_1 \varepsilon_{t-1} - \ldots - \Theta_q \varepsilon_{t-q} = \mu + \Theta(L)\varepsilon_t \tag{1}$$

where μ is an $(n \times 1)$ vector, Θ_i are $(n \times n)$ matrices and $\Theta(L) = I_n - \Theta_1 L - \ldots - \Theta_q L^q$. VMA($q$) processes are **weakly stationary** if the covariance matrix Σ of ε_t exists.

Using the fact that ε_t has no serial correlation, we have

- $\mu = \mathbb{E}[r_t]$
- $Cov(r_t, \varepsilon_t) = \Sigma$
- $\Gamma_0 = \Sigma + \Theta_1 \Sigma \Theta_1' + \ldots + \Theta_q \Sigma \Theta_q'$
- $\Gamma_k = \sum_{i=k}^{q} \Theta_i \Sigma \Theta_{i-k+1}'$ if $1 \leq k \leq q$ with $\Theta_0 = I_n$
- $\Gamma_k = 0$ and $\rho_k = 0$ for $k > q$, so that the sample cross-correlation matrices can be used to identify the order of a VMA process.

VMA(q) Model

Consider the bivariate MA(1) model

$$\begin{pmatrix} r_{1,t} \\ r_{2,t} \end{pmatrix} = \begin{pmatrix} \mu_1 \\ \mu_2 \end{pmatrix} + \begin{pmatrix} \varepsilon_{1,t} \\ \varepsilon_{2,t} \end{pmatrix} + \begin{pmatrix} \theta_{11} & \theta_{12} \\ \theta_{21} & \theta_{22} \end{pmatrix} \begin{pmatrix} \varepsilon_{1,t-1} \\ \varepsilon_{2,t-1} \end{pmatrix} \tag{2}$$

meaning that the return series at time t only depend on the current and past shocks. This is therefore a finite-memory model.

The off-diagonal elements of Θ thus show the dynamic dependence between the component series.

VMA(q) Model

We have therefore the following relationships:

- $r_{1,t}$ and $r_{2,t}$ are uncoupled series if $\theta_{12} = \theta_{21} = 0$
- There is a unidirectional dynamic relationship from $r_{1,t}$ to $r_{2,t}$ if $\theta_{12} = 0$ but $\theta_{21} \neq 0$.
- The opposite unidirectional dynamic relationship holds if $\theta_{21} = 0$ but $\theta_{12} \neq 0$.
- There is a feedback relationship between $r_{1,t}$ and $r_{2,t}$ if $\theta_{12} \neq 0$ and $\theta_{21} \neq 0$.

Estimation

Unlike the VAR model, the estimation of VMA models is much more involved, because there is no analytical solution and only numerical methods are available.

For the ML approach, there are two methods:

1. The **conditional ML** method that assumes that $\varepsilon_t = 0$ for $t \leq 0$.
2. The **exact ML** method that treats ε_t for $t \leq 0$ as additional parameters.

When the number of observations T is large, the two methods give the same estimates. Since T is generally large in finance, we focus on the conditional ML method.

VMA(q) Model

Estimation

We consider a VMA(1) model and assume that $\varepsilon_0 = 0$. Under such assumption and rewriting the model as $\varepsilon_t = r_t - \mu - \Theta_1 \varepsilon_{t-1}$, we compute the shock ε_t recursively as

$$\varepsilon_1 = r_1 - \mu \tag{3}$$

$$\varepsilon_2 = r_2 - \mu - \Theta_1 \varepsilon_1 \tag{4}$$

$$\dots \tag{5}$$

So, the likelihood function of the data becomes

$$f(r_1, ..., r_T | \mu, \Theta_1, \Sigma) = \prod_{\left(-\frac{1}{2}\varepsilon_t' \Sigma^{-1} \varepsilon_t\right)} \frac{1}{(2\pi)^{n/2} \left|\Sigma\right|^{1/2}} \exp\left(-\frac{1}{2}\epsilon_t^\top \Sigma^{-1} \epsilon_t\right) \tag{6}$$

This expression can be maximized to obtain the parameter estimates.

VARMA(p,q) Model

A vector autoregressive moving-average model of order p and q, or VARMA(p,q), is written as

$$\begin{aligned} r_t &= \mu + \sum_{i=1}^{p} \Phi_i r_{t-i} + \varepsilon_t + \sum_{i=1}^{q} \Theta_i \varepsilon_{t-i} \\ \Phi(L) r_t &= \mu + \Theta(L) \varepsilon_t \end{aligned} \tag{7}$$

where Φ_i and Θ_i are $(n \times n)$ matrices such that $\Phi(L) = I_n - \Phi_1 L - ... - \Phi_p L^p$ and$\Theta(L) = I_n - \Theta_1 L - ... - \Theta_q L^q$, and μ is an $(n \times 1)$ vector.

The generalization of ARMA models to the multivariate case encounters some new issues that do not occur in developing VAR and VMA models.

VARMA(p,q) Model

The main problem with VARMA is **identifiability**. Unlike the univariate ARMA models, VARMA models may not be uniquely defined.
For instance the VMA(1) model

$$\begin{pmatrix} r_{1,t} \\ r_{2,t} \end{pmatrix} = \begin{pmatrix} \varepsilon_{1,t} \\ \varepsilon_{2,t} \end{pmatrix} + \begin{pmatrix} 0 & -2 \\ 0 & 0 \end{pmatrix} \begin{pmatrix} \varepsilon_{1,t-1} \\ \varepsilon_{2,t-1} \end{pmatrix} \tag{8}$$

is identical to the VAR(1) model

$$\begin{pmatrix} r_{1,t} \\ r_{2,t} \end{pmatrix} - \begin{pmatrix} 0 & -2 \\ 0 & 0 \end{pmatrix} \begin{pmatrix} r_{1,t-1} \\ r_{2,t-1} \end{pmatrix} = \begin{pmatrix} \varepsilon_{1,t} \\ \varepsilon_{2,t} \end{pmatrix} \tag{9}$$

VARMA(p,q) Model

The equivalence of the two models is easy to show since we have

VMA model	VAR model
$r_{1,t} = \varepsilon_{1,t} - 2\varepsilon_{2,t-1}$	$r_{1,t} + 2r_{2,t-1} = \varepsilon_{1,t}$
$r_{2,t} = \varepsilon_{2,t}$	$r_{2,t} = \varepsilon_{2,t}$

In general VAR and VMA models are sufficient in most financial applications to capture the statistical features of the data.

Causality Tests

Test for Granger Causality

Granger causality is defined in terms of linear predictions.

Definition: Y_t does not **Granger cause** X_t iif the best linear prediction of X_t given X_{t-1} and Y_{t-1} does not depend on Y_{t-1}. More formally, let $\Omega_{t-1}(X)$ be the information set containing all the lags of X: $\Omega_{t-1}(X) = (X_\tau, \tau \leq t-1)$. Now, we say Y_t does not Granger cause X_t iif

$$MSE[X_t|\Omega_{t-1}(X), \Omega_{t-1}(Y)] = MSE[X_t|\Omega_{t-1}(X)] \qquad \forall t \tag{10}$$

where MSE is the mean squared error

$$MSE[X_t|\Omega_{t-1}(X)] = \frac{1}{T}\sum_{t=1}^{T} \hat{\varepsilon}^2_{X,t} \tag{11}$$

With $\hat{\varepsilon}_{X,t} = X_t - EL[X_t|\Omega_{t-1}(X)]$ where EL denotes linear regression.

Test for Granger Causality

The test of Granger causality is very simple to implement in a VAR model (see Geweke, 1982). Consider the three following representations:

- **The VAR(p) model:**

$$\begin{aligned} Y_t &= \Phi_{11}(L)Y_t + \Phi_{12}(L)X_t + \varepsilon_{1,t} \\ X_t &= \Phi_{22}(L)X_t + \Phi_{21}(L)Y_t + \varepsilon_{2,t} \end{aligned} \tag{12}$$

  with $V\begin{bmatrix} \varepsilon_{1,t} \\ \varepsilon_{2,t} \end{bmatrix} = \Sigma$

- **The joint dynamics** of Y_t and X_t assuming no cross-correlation between the error terms:
  $Y_t = M_{11}(L)Y_t + M_{12}(L)X_t + \eta_{1,t}$ with $V[\eta_{1,t}] = \Sigma_{12}$
  $X_t = M_{22}(L)X_t + M_{21}(L)Y_t + \eta_{2,t}$ with $V[\eta_{2,t}] = \Sigma_{21}$
- **The marginal dynamics** of Y_t and X_t as function of their own lags only: $Y_t = m_1(L)Y_t + \nu_{1,t}$ with $V[\nu_{1,t}] = \Sigma_1$
  $X_t = m_2(L)X_t + \nu_{2,t}$ with $V[\nu_{2,t}] = \Sigma_2$

Test for Granger Causality - The non-causality hypotheses and the causality measures

- The test statistic for the null hypothesis that Y does not Granger cause X ($H_{Y \to X}$) is

$$C_{Y \to X} = \log \left(\frac{|\Sigma_2|}{|\Sigma_{21}|} \right) \tag{13}$$

- The test statistic for the null hypothesis that X does not Granger cause Y ($H_{X \to Y}$) is

$$C_{X \to Y} = \log \left(\frac{|\Sigma_1|}{|\Sigma_{12}|} \right) \tag{14}$$

- The test statistic for the null hypothesis that there is no instantaneous relationship between X and Y ($H_{Y \leftrightarrow X}$) is

$$C_{X \leftrightarrow Y} = \log \left(\frac{|\Sigma_{12}||\Sigma_{21}|}{|\Sigma|} \right). \tag{15}$$

Test for Granger Causality - The non-causality hypotheses and the causality measures

The test statistic for the null hypothesis that there is no interdependence between Y and X is

$$C_{Y,X} = \log\left(\frac{|\Sigma_1||\Sigma_2|}{|\Sigma|}\right) = C_{Y\to X} + C_{X\to Y} + C_{X\leftrightarrow Y}. \tag{16}$$

Empirical counterparts to the covariance elements need to be used to obtain the test statistics.

Then under the null hypothesis, the asymptotic distributions for the four measures of causality are (Geweke, 1982):

$$\begin{aligned} T \times C_{Y\to X} &\overset{a}{\sim} \chi^2(p) \\ T \times C_{Y\leftrightarrow X} &\overset{a}{\sim} \chi^2(1) \end{aligned} \tag{17}$$

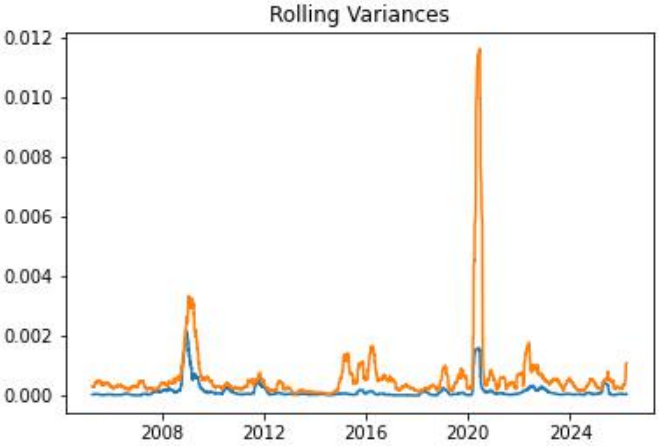
$$\begin{aligned} T \times C_{X\to Y} &\overset{a}{\sim} \chi^2(p) \\ T \times C_{Y,X} &\overset{a}{\sim} \chi^2(2p+1) \end{aligned} \tag{18}$$

Practical Application - Oil Volatility Shocks Transmission

Objective

- Test whether WTI volatility predicts S&P500 volatility
- Method: Geweke (1982) causality measures
- Focus:
  - WTI $\rightarrow$ SP500
  - SP500 $\rightarrow$ WTI
  - Instantaneous link

Variance Series



Full Sample Results

Test	Statistic	p-value
WTI $\rightarrow$ SP500	70.484292	4.440892e-16
SP500 $\rightarrow$ WTI	100.417637	0.000000e+00
Instantaneous	114.484010	0.000000e+00
Total	285.385939	0.000000e+00

Subsample Results

Period	Stat	p-value
2005-2009	-861.241362	1.0
2010-2014	1941.182262	0.0
2015-2019	2910.520827	0.0
2020-2024	-432.389797	1.0

Impulse Response Functions

Impulse Responses

Impulse responses are based on the Wold decomposition of a multivariate model:

$$r_t = \varepsilon_t + \Psi_1 \varepsilon_{t-1} + \Psi_2 \varepsilon_{t-2} + ... = \Psi(L)\varepsilon_t \tag{19}$$

where the matrix Ψ_s has the interpretation $\Psi_s = \frac{\partial r_{t+s}}{\partial \varepsilon_t'}$

Impulse Responses

The element (i,j) of matrix Ψ_s is the impact of a one-unit increase in the jth variable's innovation at date t ($\varepsilon_{j,t}$) for the value of the ith variable at time $t+s$ ($r_{i,t+s}$), when all other innovations at all dates are constant.

If we assume that the first element of ε_t is changed by $\delta_1, \ldots,$ the nth element of ε_t by δ_n, then the combined effect of these changes on the value of the vector r_{t+s} is given by

$$\Delta r_{t+s} = \frac{\partial r_{t+s}}{\partial \varepsilon_{1,t}}\delta_1 + \frac{\partial r_{t+s}}{\partial \varepsilon_{2,t}}\delta_2 + \ldots + \frac{\partial r_{t+s}}{\partial \varepsilon_{n,t}}\delta_n = \Psi_s \delta \tag{20}$$

A plot of element (i,j) of Ψ_s as a function of s is the **impulse response function**.

Impulse Responses: Case of VAR(1) Model

Assume a VAR(1) model

$$r_t = \Phi_1 r_{t-1} + \varepsilon_t \tag{21}$$

where Φ_1 is an $(n \times n)$ matrix and $\varepsilon_t \sim (0, \Sigma)$.

The relation between r_t and a past shocks is given by

$$r_t = \Phi_1(\Phi_1 r_{t-2} + \varepsilon_{t-1}) + \varepsilon_t = \varepsilon_t + \Phi_1 \varepsilon_{t-1} + \Phi_1^2 \varepsilon_{t-2} + ... \tag{22}$$

Therefore, the impact of a one-unit increase in innovations at date t (ε_t) for the value of the r_{t+s} is Φ_1^s.

Remark: The impulse response is another kind of causality measure, although it is different from Granger causality.

Impulse Responses: Example

Consider the following model

$$\begin{pmatrix} r_{1,t} \\ r_{2,t} \\ r_{3,t} \end{pmatrix} = \begin{pmatrix} 0.5 & 0 & 0 \\ 0.1 & 0.1 & 0.3 \\ 0 & 0.2 & 0.3 \end{pmatrix} \begin{pmatrix} r_{1,t-1} \\ r_{2,t-1} \\ r_{3,t-1} \end{pmatrix} + \begin{pmatrix} \varepsilon_{1,t} \\ \varepsilon_{2,t} \\ \varepsilon_{3,t} \end{pmatrix} \tag{23}$$

or $r_t = \breve{}_1 r_{t-1} + \varepsilon_t$.

Assume a unit shock in the first variable at time $t = 0$:

$$r_0 = \begin{pmatrix} r_{1,0} \\ r_{2,0} \\ r_{3,0} \end{pmatrix} = \begin{pmatrix} \varepsilon_{1,0} \\ \varepsilon_{2,0} \\ \varepsilon_{3,0} \end{pmatrix} = \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix} \tag{24}$$

Impulse Responses: Example

Then at time $t = 1$, we have:

$$r_1 = \begin{pmatrix} r_{1,1} \\ r_{2,1} \\ r_{3,1} \end{pmatrix} = \breve{}_1 r_0 = \begin{pmatrix} 0.5 & 0 & 0 \\ 0.1 & 0.1 & 0.3 \\ 0 & 0.2 & 0.3 \end{pmatrix} \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix} = \begin{pmatrix} 0.5 \\ 0.1 \\ 0 \end{pmatrix} \tag{25}$$

At time $t = 2$, we have:

$$r_2 = \begin{pmatrix} r_{1,2} \\ r_{2,2} \\ r_{3,2} \end{pmatrix} = \breve{}_1 r_1 = \begin{pmatrix} 0.5 & 0 & 0 \\ 0.1 & 0.1 & 0.3 \\ 0 & 0.2 & 0.3 \end{pmatrix} \begin{pmatrix} 0.5 \\ 0.1 \\ 0 \end{pmatrix} = \begin{pmatrix} 0.25 \\ 0.06 \\ 0.02 \end{pmatrix} \tag{26}$$

Impulse Responses: Example

In the previous cases, we have assumed a shock in only one variable at a time. This assumption is reasonable if shocks are independent.

When shocks are not independent, which is likely as $\varepsilon_t \sim (0, \Sigma)$, we need to orthogonalize them. We consider the following Cholesky decomposition: $\Sigma = PP'$ where P is a lower triangular matrix. Then we can transform

$$r_t = \varepsilon_t + \Psi_1 \varepsilon_{t-1} + \Psi_2 \varepsilon_{t-2} + ... = \Psi(L)\varepsilon_t \tag{27}$$

as follows

$$r_t = \Gamma_0 w_t + \Gamma_1 w_{t-1} + \Gamma_2 w_{t-2} + ... = \Gamma(L) w_t \tag{28}$$

where $\Gamma_s = \Psi_s P$, $w_t = P^{-1}\varepsilon_t$ and $V[w_t] = I_n$.

Matrices Γ_s correspond to **orthogonalized impulse responses**.

Impulse Response Functions

Study of the impact of a shock in one variable over a set of variables.
Shocks are captured through ϵ_t and their instantaneous impact through $\sqrt{\Sigma}$.

Assume a VAR(1) with two variables:

$$X_t = \Phi_0 + \Phi_1 X_{t-1} + \sqrt{\Sigma}\epsilon_t, \tag{29}$$

with $\epsilon_t = \left(\epsilon_t^1 \ \epsilon_t^2\right)$.
Assume Σ has the following form:

$$\Sigma = \begin{bmatrix} \sigma_1^2 & 0 \\ 0 & \sigma_2^2 \end{bmatrix} \tag{30}$$

With that specification, shocks are instantaneously *orthogonal*, meaning shocks in variable 1 do no influence the shocks in variable 2.

Impulse Response Functions

Now assume Σ is as follows:

$$\Sigma = \begin{bmatrix} \sigma_1^2 & \sigma_{12} \\ \sigma_{12} & \sigma_2^2 \end{bmatrix} \tag{31}$$

Assume we have a unit shock on variable 1 and that

$$\sqrt{\Sigma} = \begin{bmatrix} a & b \\ b & c \end{bmatrix}. \tag{32}$$

Impulse Response Functions

The actual shock on variable 2 writes:

$$x_t^{(2)} = \Phi_0^{(2)} + \Phi_1^{(12)} x_{t-1}^{(1)} + \Phi_1^{(22)} x_{t-1}^{(2)} + \underbrace{b\epsilon_t^{(1)} + c\epsilon_t^{(2)}}_{\text{Variable 2 shock}} \tag{33}$$

Shocks for one variable are therefore a cocktail of the orthogonal shocks contained in ϵ_t.
How can we simulate shocks and their dynamic impact of the economy?

Impulse Response Functions

VAR models are *stationary* models.
Their Wold decomposition writes:

$$X_t = \mu + \Psi_0 \epsilon_t + \Psi_1 \epsilon_{t-1} + \Psi_2 \epsilon_{t-2} + \dots \tag{34}$$

With this representation in mind, it is obvious that

$$\frac{\partial X_t}{\partial \epsilon_{t-i}} = \Psi_i. \tag{35}$$

Ψ_i measure the influence of ϵ_{t-i} on X_t.

Impulse Response Functions

We need to orthogonalize the shocks. Any definite positive matrix Σ can be decomposed as follows

$$\Sigma = PP^{\top}, \tag{36}$$

where

$$P = \begin{bmatrix} a_{11} & 0 & \dots & 0 \\ a_{21} & a_{22} & \dots & 0 \\ \vdots & \vdots & \vdots & \vdots \\ a_{n1} & a_{n2} & \dots & a_{nn} \end{bmatrix}. \tag{37}$$

P is an invertible matrix.

Impulse Response Functions

We can re-write all shocks as

$$u_t = P^{-1}\epsilon_t. \tag{38}$$

With that in mind:

$$\mathbb{E}[u_t] = \mathbb{E}[P^{-1}\epsilon_t] = P^{-1}\mathbb{E}[\epsilon_t] = 0 \tag{39}$$

$$\mathbb{V}[u_t] = \mathbb{E}\left[u_t u_t^\top\right] = \mathbb{E}\left[P^{-1}\epsilon_t\epsilon_t^\top(P^{-1})^\top\right] = P^{-1}PP^\top(P^\top)^{-1} = I. \tag{40}$$

u_t is therefore an orthogonalized set of shocks. As

$$\epsilon_t = Pu_t, \tag{41}$$

we can rewrite the Wold decomposition as follows:

$$X_t = \mu + \Psi_0 P u_t + \Psi_1 P u_{t-1} + \Psi_2 P u_{t-2} + \dots \tag{42}$$

Shocks in u_t are now orthogonal.

Impulse Response Functions

The system of shocks can be re-written as follows:

$$\begin{bmatrix} \epsilon_{1t} \\ \epsilon_{2t} \\ \epsilon_{3t} \\ \vdots \end{bmatrix} = \begin{bmatrix} P_{11} & 0 & 0 & \dots \\ P_{21} & P_{22} & 0 & \dots \\ P_{31} & P_{32} & P_{33} & \dots \\ \vdots & \vdots & \vdots & \vdots \end{bmatrix} \begin{bmatrix} u_{1t} \\ u_{2t} \\ u_{3t} \\ \vdots \end{bmatrix}. \tag{43}$$

This implies that

$$\epsilon_{1t} = P_{11} u_{1t} \tag{44}$$

$$\epsilon_{2t} = P_{21} u_{1t} + P_{22} u_{2t} \tag{45}$$

$$\epsilon_{3t} = P_{31} u_{1t} + P_{32} u_{2t} + P_{33} u_{3t}. \tag{46}$$

The way variables are ordered is thefore important: the first variable will be influenced by coincident shocks coming from other variables.

Impulse Response Functions

Another way of seeing it:

$$\mathbb{V}[\epsilon_{1t}] = \mathbb{V}[P_{11}u_{1t}] = P_{11}^2 \tag{47}$$

$$\mathbb{V}[\epsilon_{2t}] = \mathbb{V}[P_{21}u_{1t} + P_{22}u_{2t}] = \mathbb{V}\left[\frac{P_{21}}{P_{11}}\epsilon_{1t} + P_{22}u_{2t}\right] \tag{48}$$

This last expression shows how the variance of ϵ_{2t} can be split between what actually comes from ϵ_{1t} and what comes from its own shocks.
As $\epsilon_{1t} \perp u_{2t}$, then

$$\mathbb{V}\left[\frac{P_{21}}{P_{11}}\epsilon_{1t} + P_{22}u_{2t}\right] = \frac{P_{21}^2}{P_{11}^2} + P_{22}^2 \tag{49}$$

$\Rightarrow$ Now, we need a way to compute P. Most common way: **Choleski decomposition**.

Impulse Response Functions

The AR(1) case:

$$
\begin{aligned}
X_t &= \Phi_0 + \Phi_1 X_{t-1} + \sqrt{\Sigma}\epsilon_t && (50)\\
&= \Phi_0 + \Phi_1 X_{t-1} + \nu_t && (51)\\
&= \Phi_0 \sum_{i=0}^{\infty} \Phi_1^i + \sum_{i=0}^{\infty} \Phi_1^i \nu_{t-i}. && (52)\\
&= \Phi_0 \sum_{i=0}^{\infty} \Phi_1^i + \sum_{i=0}^{\infty} \Phi_1^i P u_{t-i}. && (53)
\end{aligned}
$$

The impulse response function is the function which for a given h computes

$$
h \mapsto \frac{\partial X_{t+h}}{\partial u_t} = \Phi_1^h P. \tag{54}
$$

Matrices $\Gamma_h 0 \Phi_1^h P$ correspond to **orthogonalized impulse responses**. Confidence intervals are usually built using bootstrap.

Practical Application - US economy Macro Model

Time for an Application Model

Let's use all of that with a more concrete example.

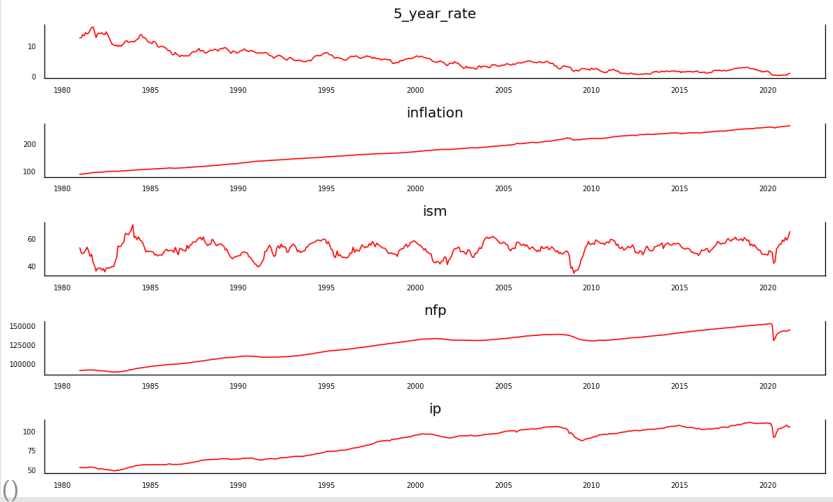
Using the following timeseries:

- the US 5-year rate
- the year-over-year change in the Consumer Price Index as a measure of inflation
- the US ISM, an indicator of economic conditions
- the non-farm payrolls, the number of job creation in the US
- the industrial production index.

We will use monthly time series, from December 1980 to March 2021.
Let's try creation a VAR model around these time series and try to understand what could be the impact of an inflation shock on the rest of the variables describing the US economy.

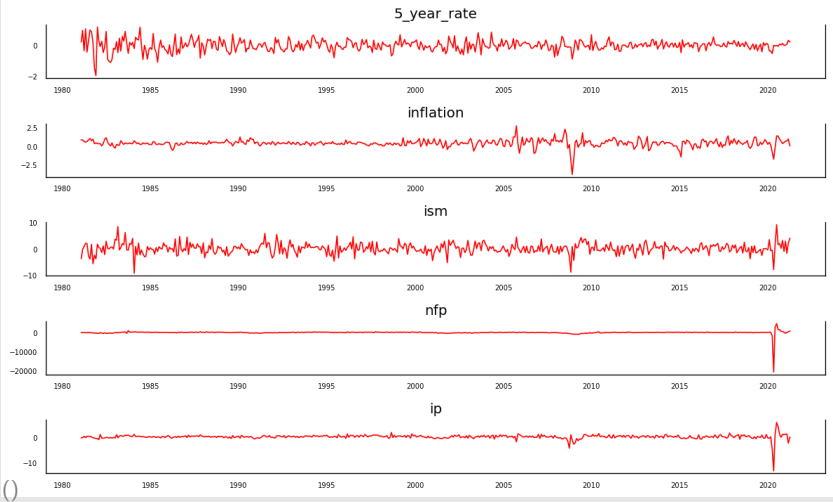
Time for an Application

First step: Let's Take a Look at the Timeseries.



Time for an Application

First step: Let's Take a Look at the Timeseries. With first order differences:



Time for an Application Model

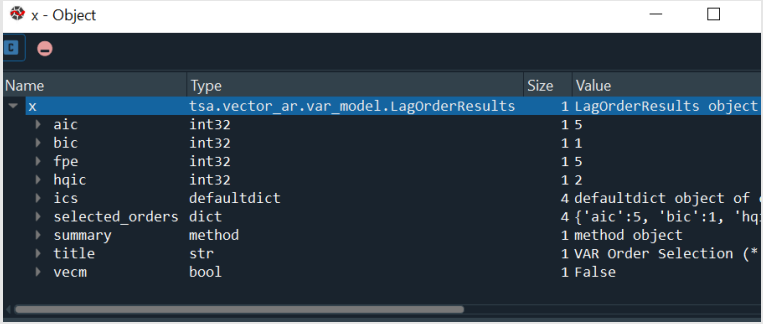
Second step: Granger causality test.

```
    ...:
    ...: test=grangers_causation_matrix(Macro_series, variables = Macro_series.columns)

In [90]: test
Out[90]:
                 5_year_rate_x  inflation_x   ism_x   nfp_x    ip_x
5_year_rate_y           1.0000       0.0019  0.0086  0.0241  0.0453
inflation_y             0.0221       1.0000  0.0152  0.4487  0.1590
ism_y                   0.0000       0.0004  1.0000  0.0000  0.0004
nfp_y                   0.0258       0.0256  0.0002  1.0000  0.0000
ip_y                    0.0007       0.0011  0.0000  0.0000  1.0000

In [91]:
```

$\Rightarrow$ These series are causing each others!

Time for an Application Model

Third step: Lag Selection.



x - Object

Name	Type	Size	Value
x	tsa.vector_ar.var_model.LagOrderResults	1	LagOrderResults object
aic	int32	1	5
bic	int32	1	1
fpe	int32	1	5
hqic	int32	1	2
ics	defaultdict	4	defaultdict object of
selected_orders	dict	4	{'aic':5, 'bic':1, 'hq
summary	method	1	method object
title	str	1	VAR Order Selection (*
vecm	bool	1	False

$\Rightarrow$ No such thing as a single answer, clearly. Let's go for 2 lags.

Time for an Application Model

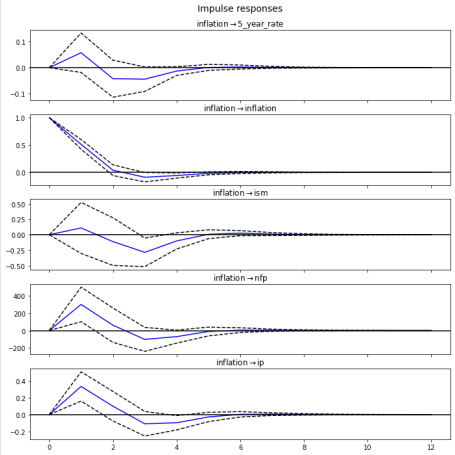
Fourth step: Parameter Estimation.

```
--------------------------------------------------------------------------------

Results for equation inflation
==============================================================================
                   coefficient       std. error           t-stat             prob
------------------------------------------------------------------------------
const                 0.262843         0.026219           10.025            0.000
L1.5_year_rate        0.080693         0.054975            1.468            0.142
L1.inflation          0.512628         0.045811           11.190            0.000
L1.ism                0.009804         0.010093            0.971            0.331
L1.nfp               -0.000078         0.000032           -2.441            0.015
L1.ip                 0.079246         0.035276            2.246            0.025
L2.5_year_rate        0.065465         0.055658            1.176            0.240
L2.inflation         -0.231696         0.045487           -5.094            0.000
L2.ism               -0.005469         0.010159           -0.538            0.590
L2.nfp               -0.000132         0.000032           -4.086            0.000
L2.ip                 0.142168         0.036936            3.849            0.000
==============================================================================
```

$\Rightarrow$ Inflation relates positively to past industrial production.

Time for an Application Model

Fifth step: Estimation of the IRFS.



$\Rightarrow$ How can we read that chart?

Variance Decomposition

We start again with the Wold decomposition of a multivariate model:

$$r_t = \varepsilon_t + \Psi_1 \varepsilon_{t-1} + \Psi_2 \varepsilon_{t-2} + ... = \Psi(L)\varepsilon_t \tag{55}$$

We are interested in the error in forecasting r_t, s periods into the future:

$$r_{t+s} - \hat{r}_{t+s/t} = \varepsilon_{t+s} + \Psi_1 \varepsilon_{t+s-1} + \Psi_2 \varepsilon_{t+s-2} + ... + \Psi_{s-1} \varepsilon_{t+1} \tag{56}$$

with

$$\hat{r}_{t+s/t} = \Psi_s \varepsilon_t + \Psi_{s+1} \varepsilon_{t-1} + \Psi_{s+2} \varepsilon_{t-2} + ... \tag{57}$$

The mean squared error of the s-period-ahead forecast is

$$MSE(\hat{r}_{t+s/t}) = E[(r_{t+s} - \hat{r}_{t+s/t})(r_{t+s} - \hat{r}_{t+s/t})'] \tag{58}$$

$$= \Sigma + \Psi_1 \Sigma \, \Psi_1' + \Psi_2 \Sigma \, \Psi_2' + ... + \Psi_{s-1} \Sigma \, \Psi_{s-1}' \tag{59}$$

Pratical Application - Building Spillover Measures

Variance Decomposition

The h-step ahead forecast error variance decomposition :

$$\theta_{ij}(h) = \sigma_{ii}^{-1} \frac{\sum_{h=0}^{h-1} (e_i' \Psi_h \Sigma e_j)^2}{\sum_{h=0}^{H-1} e_i' \Psi_h \Sigma (\Psi_h)' e_j}, \tag{60}$$

e_i a vector of zeros but for its ith element equal to 1.

Rescaled quantities:

$$\tilde{\theta}_{ij}(H) = \frac{\theta_{ij}(H)}{\sum_{j=1}^{N} \theta_{ij}(H)}, \Rightarrow \sum_{j=1}^{N} \tilde{\theta}_{ij}(H) = 1 \tag{61}$$

$\tilde{\theta}_{ij}(H)$ are "**variance shares**": measure of the share of the total forecasting error variance for series i that is explained through series j.

Variance Decomposition

- **Spillover index**: $S(H)$ measure of how much the series considered are receiving shocks that are not theirs:

$$S(H) = \frac{\sum_{i,j=1,i\neq j}^{N} \tilde{\theta}_{ij}(H)}{N} \times 100. \tag{62}$$

- **Spillover received**:

$$S_{i\leftarrow}(H) = \frac{\sum_{j=1,j\neq i}^{N} \tilde{\theta}_{ij}(H)}{\sum_{j=1}^{N} \tilde{\theta}_{ij}(H)} \times 100. \tag{63}$$

- **Spillover given**:

$$S_{i\rightarrow}(H) = \frac{\sum_{j=1,j\neq i}^{N} \tilde{\theta}_{ji}(H)}{\sum_{j=1}^{N} \tilde{\theta}_{ji}(H)} \times 100. \tag{64}$$

- **Net spillovers**:

$$S_i(H) = S_{i\rightarrow}(H) - S_{i\leftarrow}(H). \tag{65}$$

Variance Decomposition

Empirical application to credit spreads:

1. When dealing with credit portfolio, **sector exposure** is a natural partition used by credit investment managers to gauge the origin of their performance.

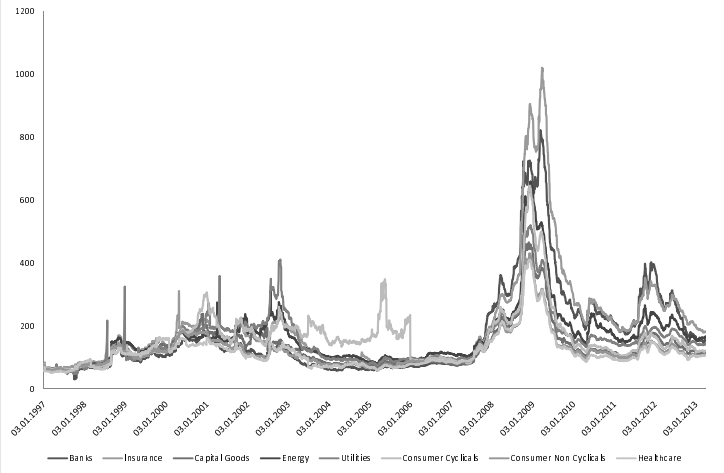
2. Selected sectors are supposed to **react more aggressively** to market evolutions than others: bonds issued by financial firm vs. pharmaceutical companies.

3. However sensitivity to market evolutions says little on **how shocks spread across sectors** / within which sector do turmoils usually start and end.

4. This paper provides empirical evidence that **not every sectors are net volatility spillover givers**.

Variance Decomposition n

Evolution of credit spreads by sector

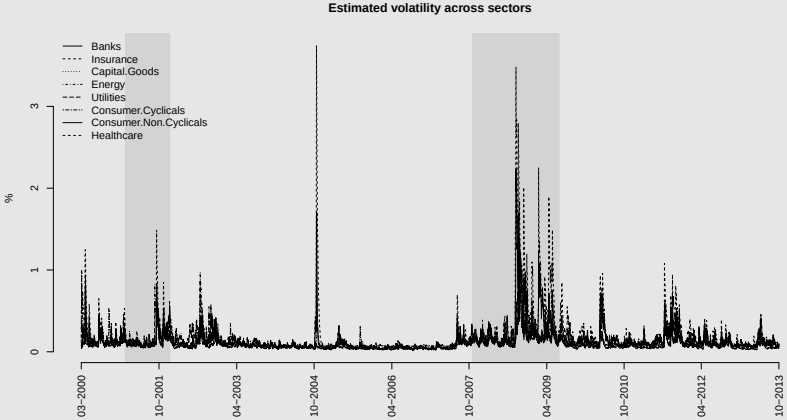


Variance Decomposition

- **Dataset**:
  - Excess returns on credit bonds, from Merill Lynch BoA.
  - US Investment Grade bonds.
  - 2001-2013.
  - By sectors: Banks, Insurance, Cap. goods, Energy, Util., Cons. cyc., Cons. non cyc., Health.
- **Based on blended volatility measures**.

Variance Decomposition

Time series estimates of each sector's volatility



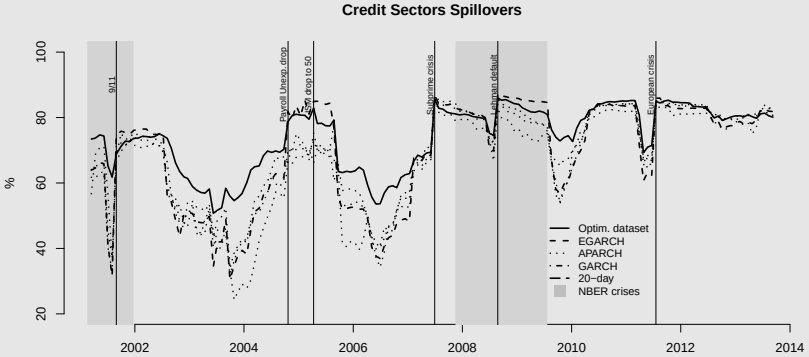
Variance Decomposition

Spillovers over the 2001-2013 period

	Banks	Ins.	Cap. Goods	Ener.	Util.	CC	CNC	Health.	Sp. Rec.
Banks	13.42	40.29	3.76	7.86	5.62	7.62	13.25	8.17	86.58
Ins.	0.47	76.77	1.17	1.05	0.74	2.65	16.79	0.36	23.23
Cap. Goods	3.09	48.94	8.68	3.92	3.52	16.5	13.19	2.16	91.32
Ener.	5.74	31.85	2.88	21.51	15.24	5.91	12.95	3.92	78.49
Util.	5.09	29.78	3.32	19.56	17.91	5.61	15.35	3.39	82.09
CC	2.85	51.04	7.67	3.98	2.95	17.69	11.62	2.19	82.31
CNC	0.67	68.68	1.27	1.37	1.26	2.5	23.75	0.5	76.25
Health.	12.38	40.08	4.02	8.11	5.59	8.99	12.01	8.81	91.19
Sp. Given	69.3	80.18	73.52	68.08	66.1	73.78	80.03	70.13	**76.43**
Net Sp.	-17.28	56.95	-17.8	-10.41	-15.99	-8.53	3.78	-21.06	

Variance Decomposition

Time series estimates of each sector's volatility



Practical Application: Inflation Shocks and Fed Rates in a VAR

Instructions

- Open the notebook `EMIF_VAR_inflation_shock_notebook_executed.ipynb`.
- Run the code cell by cell and read the comments before moving to the next block.
- Focus on four questions:
  - How do we choose the VAR lag length?
  - How do we interpret the coefficients and diagnostics?
  - What happens after a one standard deviation inflation shock?
  - Why did a model estimated at the end of 2021 fail to fully anticipate 2022?
- The exported charts and tables are used in the slides so that you can present the key messages in class.

Economic setup

We study the monthly interaction between inflation, Fed rates, and capacity utilization.

The reduced-form VAR with p lags is:

$$y_t = c + A_1 y_{t-1} + A_2 y_{t-2} + \cdots + A_p y_{t-p} + u_t \tag{66}$$

with:

$$y_t = \begin{pmatrix} \text{Inflation}_t \\ \text{Fed Rates}_t \\ \text{Capacity Utilization Rate}_t \end{pmatrix} \tag{67}$$

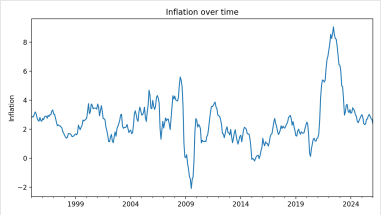
The ordering matters for the shock analysis. Inflation is placed first so that a contemporaneous inflation shock can affect the other variables within the same month.

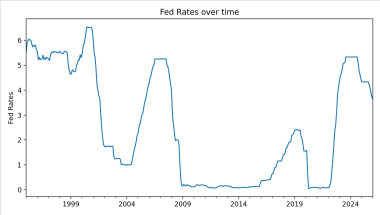
Summary statistics

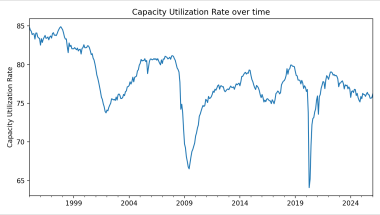
Summary statistics

	mean	std	min	max
Inflation	2.5481	1.6066	-2.0972	9.0597
Fed Rates	2.5552	2.2395	0.05	6.54
Capacity Utilization Rate	77.9678	3.6083	64.0847	84.8918

The raw data


Inflation over time
Inflation
8
6
4
2
0
−2
1999
2004
2009
2014
2019
2024


Fed Rates over time
Fed Rates
6
5
4
3
2
1
0
1999
2004
2009
2014
2019
2024


Capacity Utilization Rate over time
Capacity Utilization Rate
85
80
75
70
65
1999
2004
2009
2014
2019
2024

Choosing the lag length

Lag selection criteria

	AIC	BIC	HQIC	FPE
0	4.4063	4.4386	4.4192	81.9667
1	-5.8876	-5.7583	-5.8362	0.0028
2	-6.6412	-6.4149	-6.5512	0.0013
3	-6.7195	-6.3963	-6.591	0.0012
4	-6.7416	-6.3215	-6.5746	0.0012
5	-6.7122	-6.1952	-6.5067	0.0012
6	-6.7182	-6.1042	-6.4741	0.0012
7	-6.7105	-5.9995	-6.4278	0.0012
8	-6.676	-5.8681	-6.3548	0.0013
9	-6.6594	-5.7545	-6.2996	0.0013
10	-6.6406	-5.6387	-6.2423	0.0013
11	-6.6718	-5.573	-6.2349	0.0013
12	-6.6451	-5.4493	-6.1697	0.0013

AIC and FPE favor a richer specification, while BIC prefers a more parsimonious one. For monthly data, a VAR(4) is a reasonable teaching choice because it captures short-run dynamics without becoming too heavy.

Selected coefficient tables

Inflation equation coefficients

	coef	std_err	t_stat	p_value
const	-0.1491	0.5804	-0.2568	0.7973
L1.Inflation	1.4255	0.0537	26.5332	0.0
L1.Fed Rates	0.43	0.1528	2.815	0.0049
L1.Capacity Utilization Rate	0.0232	0.0275	0.8436	0.3989
L2.Inflation	-0.653	0.093	-7.019	0.0
L2.Fed Rates	-0.6866	0.2783	-2.4675	0.0136
L2.Capacity Utilization Rate	0.0137	0.0404	0.3388	0.7348
L3.Inflation	0.249	0.0928	2.6828	0.0073
L3.Fed Rates	-0.0067	0.2852	-0.0233	0.9814
L3.Capacity Utilization Rate	0.0113	0.0393	0.2879	0.7734
L4.Inflation	-0.065	0.0539	-1.2055	0.228
L4.Fed Rates	0.2616	0.1557	1.6805	0.0929
L4.Capacity Utilization Rate	-0.0447	0.0265	-1.6864	0.0917

Fed rates equation coefficients

	coef	std_err	t_stat	p_value
const	-0.2405	0.2027	-1.1864	0.2355
L1.Inflation	-0.0229	0.0188	-1.2187	0.223
L1.Fed Rates	1.5461	0.0533	28.9819	0.0
L1.Capacity Utilization Rate	0.0141	0.0096	1.4747	0.1403
L2.Inflation	0.0493	0.0325	1.5184	0.1289
L2.Fed Rates	-0.5358	0.0972	-5.5137	0.0
L2.Capacity Utilization Rate	-0.0231	0.0141	-1.6417	0.1007
L3.Inflation	-0.0682	0.0324	-2.1043	0.0354
L3.Fed Rates	0.1035	0.0996	1.039	0.2988
L3.Capacity Utilization Rate	0.0247	0.0137	1.8002	0.0718
L4.Inflation	0.0501	0.0188	2.6641	0.0077
L4.Fed Rates	-0.1265	0.0544	-2.3281	0.0199
L4.Capacity Utilization Rate	-0.0125	0.0093	-1.3467	0.1781

Identification of the inflation shock

For impulse responses, we use an orthogonalized innovation:

$$u_t = P\varepsilon_t \tag{68}$$

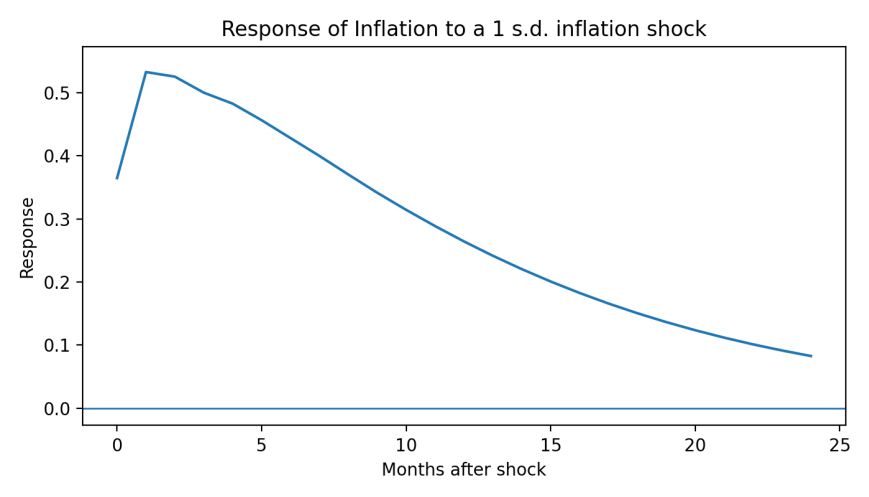
with:

$$\Sigma_u = PP' \tag{69}$$

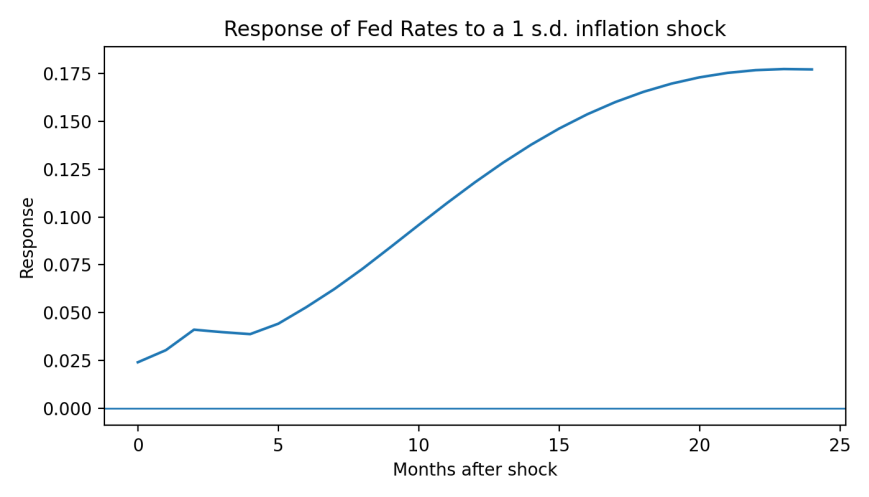
A one standard deviation inflation shock is then traced through the system. The interpretation should focus on:

- the sign of the response,
- the magnitude of the response,
- the persistence of the response.

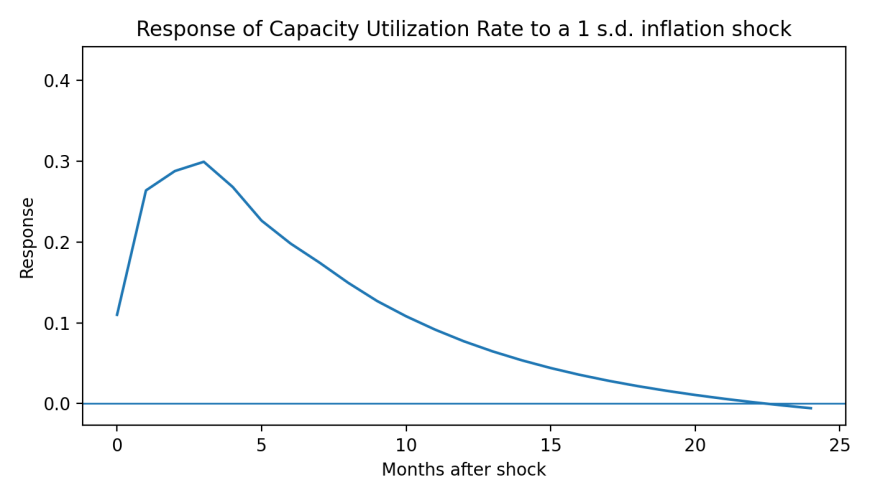
Impulse response of inflation



Impulse response of Fed rates



Impulse response of capacity utilization



Forecasting from the end of 2021

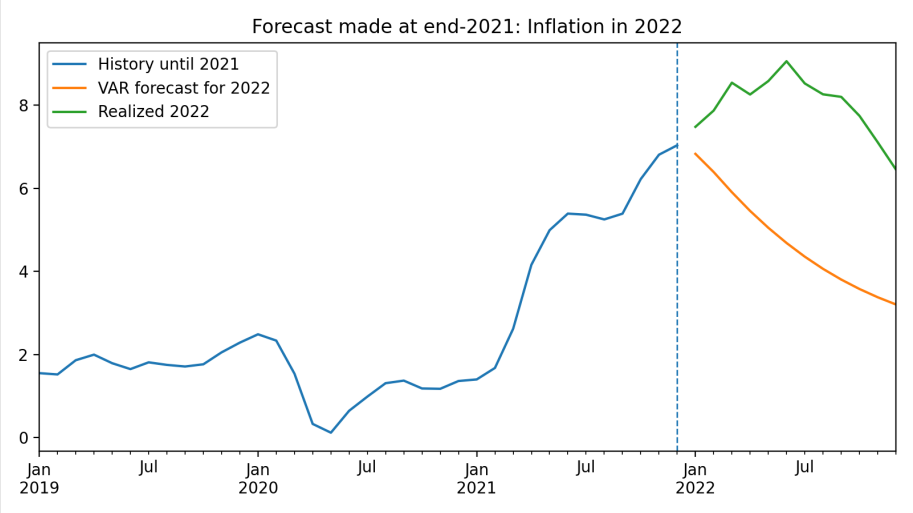
We estimate the VAR only on information available up to December 2021 and forecast the twelve months of 2022.

The key comparison is:

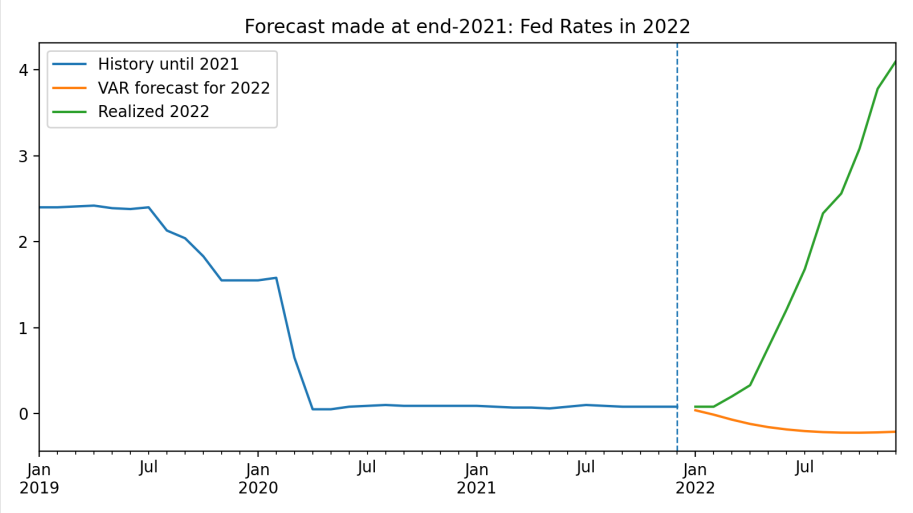
$$\text{Forecast made at end-2021} \quad \text{vs} \quad \text{Realized values in 2022} \tag{70}$$

If 2022 inflation and Fed rates are much higher than forecast, this is evidence of a large inflation shock that was not already embedded in the historical dynamics of the system.

Inflation: forecast versus realized 2022



Fed rates: forecast versus realized 2022



Forecast errors in 2022

Forecast vs realized averages in 2022

	0
Forecast average 2022 inflation	4.7253
Realized average 2022 inflation	8.0077
Forecast average 2022 Fed rates	-0.15
Realized average 2022 Fed rates	1.6833

2022 forecast RMSE

	RMSE
Inflation	3.479
Fed Rates	2.3512
Capacity Utilization Rate	1.8062

The miss in 2022 is not just a forecasting failure. It is a useful economic lesson: when inflation moves outside the range suggested by normal historical dynamics, policy rates must also adjust by more than the model had anticipated.

Takeaways

- A VAR is a convenient framework to study the joint dynamics of inflation, policy rates, and activity.
- Identification matters: the ordering of variables affects orthogonalized impulse responses.
- A one standard deviation inflation shock can be traced through the whole system with impulse responses.
- Forecast errors are economically informative: the gap between the end-2021 forecast and realized 2022 values illustrates the effect of an unusual inflation shock.
- The exercise shows both the usefulness and the limits of historical models in macro-finance.